Preliminaries for UCRL

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Suppose $X_i \in \{1, 2, \ldots, S\}$ are i.i.d. r.v.'s with $p_s = P(X_i = s)$

Then, p_s can be estimated as $\underbrace{\frac{I_{X_1=s} + \cdots + I_{X_n=s}}{n}}_{\hat{p}_s}$

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} I_{X_i=s} - p_s\right| \geq \varepsilon\right)$$

$$\leq 2\exp\left(-2n\varepsilon^2\right) \quad \left(\text{by the Hoeffding bound}\right)$$

But about

$$P\left(\|\hat{p} - p\|_1 \geq \varepsilon\right) ?$$

Note that

$$\|\hat{p} - p\|_1 = \sum_{s=1}^{S} |\hat{p}_s - p_s|$$

$$= \max_{z \in \{+1,-1\}^S} \sum_{s=1}^{S} (\hat{p}_s - p_s) z_s$$

$$\Rightarrow P\left(\|\hat{p} - p\|_1 \geq \varepsilon\right)$$

$$= P\left(\max_{z \in \{+1,-1\}^S} \sum_{s=1}^{S} (\hat{p}_s - p_s) z_s \geq \varepsilon\right)$$

$$\leq \sum_{z \in \{+1,-1\}^S} P\left(\sum_{s=1}^{S} (\hat{p}_s - p_s) z_s \geq \varepsilon\right)$$

$$= P\left(\frac{1}{n}\sum_{i=1}^{n} \left(\sum_{s=1}^{S} I_{X_i=s} z_s\right) - \sum_{s=1}^{S} p_s z_s \geq \varepsilon\right)$$

$$= \sum_{z \in \{+1,-1\}^S} P\left(\frac{1}{n} \sum_{i=1}^{n} \sum_s \left(\mathbb{I}_{X_i = s} - p_s \right) z_s \geq \varepsilon \right)$$

$$\left(\text{since } \left| \sum_s \mathbb{I}_{X_i = s} z_s \right| \leq 1 \right)$$

$$\leq \sum_{z \in \{+1,-1\}^S} \exp\left(- \frac{n\varepsilon^2}{2} \right)$$

$$= 2^S \exp\left(- \frac{n\varepsilon^2}{2} \right)$$

To be clear that we used n samples, Let's denote $\hat{p}$ by $\hat{p}(n)$. So we have

$$P\left(\| \hat{p}(n) - p \|_1 \geq \varepsilon \right) \leq \underbrace{2^S \exp\left(- \frac{n\varepsilon^2}{2} \right)}_{\text{call this } \delta}$$

$$\Rightarrow \quad 2^S \exp\left(- \frac{n\varepsilon^2}{2} \right) = \delta$$

$$S \ln 2 - \frac{n\varepsilon^2}{2} = \ln \delta$$

$$\frac{n\varepsilon^2}{2} = S \ln 2 + \ln\left(\frac{1}{\delta} \right)$$

$$\text{or} \quad \varepsilon = \sqrt{\frac{2}{n}\left(S \ln 2 + \ln\left(\frac{1}{\delta} \right) \right)}$$

Therefore,

$$P\left(\| \hat{p}(n) - p \|_1 \geq \sqrt{\frac{2}{n}\left(S \ln 2 + \ln\left(\frac{1}{\delta} \right) \right)} \right) \geq \delta$$

Now, suppose we want

Now, suppose we want

$$P\left(\sup_{n\geq 1} \|\hat{p}(n) - p\|_1 \geq g(\delta)\right) \leq \delta,$$

how should we choose $g(\delta)$? Note that

$$P\left(\|\hat{p}(n) - p\|_1 \geq \sqrt{\frac{2}{n}\left(5\ln 2 + \ln \frac{n(n+1)}{\delta}\right)}\right)$$

$$\leq \frac{\delta}{n(n+1)}$$

$$\Rightarrow P\left(\sup_{n\geq 1} \|\hat{p}(n) - p\|_1 \geq \sqrt{\frac{2}{n}\left(5\ln 2 + \ln \frac{n(n+1)}{\delta}\right)}\right)$$

$$\leq \delta \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \delta \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$= \delta$$

In other words w.p. $\geq 1-\delta$,

$$\|\hat{p}(n) - p\|_1 < \sqrt{\frac{2}{n}\left(5\ln 2 + \ln \frac{n(n+1)}{\delta}\right)} \quad \forall n$$

□

Martingales

$X_1, X_2, \ldots$, be a sequence of r.v.'s is a martingale w.r.t. $Y_1, Y_2, \ldots$, if

(A) $E(X_{n+1} \mid Y_1, Y_2, \ldots, Y_n) = X_n$ and

(A) $E(X_{n+1} \mid Y_1, Y_2, .., Y_n) = X_n$ and

(B) $E(|X_n|) < \infty \quad \forall n$

Note: (i) $E(|X_n|) < \infty$ is the same as $E(X_n) = \infty$ since $E(X_n)$ is defined to be $E(X_n I_{X_n \geq 0}) + E(X_n I_{X_n < 0})$ and both of these should exist.

(ii) $E(X_{n+1} \mid Y_1, Y_2, .., Y_n)$ is a function of $Y_1, .., Y_n$. So (A) $\Rightarrow$ X_n is completely determined by $Y_1, .., Y_n$ w.p.1.

- Note: Suppose $Z_1, Z_2, ..$ is an independent seq and let $X_n = \sum_{i=1}^{n} (Z_i - E(Z_i))$

If $E(Z_i) < \infty \Rightarrow E(X_n) < \infty \quad \forall n$

$$E(X_{n+1} \mid Z_1, .., Z_n) = E\left(\sum_{i=1}^{n} (Z_i - E(Z_i)) \mid Z_1, .., Z_n\right)$$

$$+ E(Z_{n+1} - E(Z_{n+1}) \mid Z_1, .., Z_n)$$

$= 0$ by independence

$$= X_n$$

$\Rightarrow$ X_n is a martingale.

Azuma-Hoeffding inequality:

If $X_1, X_2, ..$ is a martingale w.r.t. $Y_1, Y_2, ..$, and $a_i \leq X_i - X_{i-1} \leq b_i$ (taking $X_0 = 0$), with $E(X_1) = 0$,

then $$P(X_n \geq \varepsilon) \leq \exp\left(\frac{-2n^2\varepsilon^2}{\quad}\right)$$

then

$$P\left(\frac{X_n}{n} \geq \varepsilon\right) \leq \exp\left(\frac{-2n^2\varepsilon^2}{\sum_{i=1}^{n}(b_i-a_i)^2}\right)$$

Thus, Azuma-Hoeffding generalizes Hoeffding. □

Useful lemma: for non-negative integers m_k,

$$\sum_{k=1}^{K} \frac{m_k}{\sqrt{1 \vee (m_1+\cdots+m_k)}} \leq 2\sqrt{m_1+\cdots+m_K}$$

Proof: if $b \geq a$,

$$\sqrt{b}-\sqrt{a} = \frac{b-a}{\sqrt{b}+\sqrt{a}} \geq \frac{b-a}{2\sqrt{b}}$$

$$\Rightarrow \sqrt{m_1+\cdots+m_k} - \sqrt{m_1+\cdots+m_{k-1}} \geq \frac{m_k}{2\sqrt{1+\cdots+m_k}}$$

$$\Rightarrow \sum_{k=1}^{K} \frac{m_k}{\sqrt{m_1+\cdots+m_k}} \leq 2\sqrt{m_1+\cdots+m_K}$$

Telescoping

This proof holds if $m_1+\cdots+m_k > 0$. If this is $=0$, then the LHS $=0$ & RHS $=0$

□

Estimating probability transitions

Estimating probability transitions

- Fix a policy π_t: suppose we obtain a trajectory of length H from π_t:

$$s_0^t, a_0^t, s_1^t, a_1^t, \ldots, s_{H-1}^t, a_{H-1}^t$$

- Let us say that we obtain trajectories from many such policies $\pi_1, \ldots, \pi_t$

- Let $n_{s,a,s'}(t)$ = total # of (s, a, s') triples we have seen in trajectories $1, 2, \ldots, t$

and $n_{s,a}(t)$ = $\ldots$ (s, a) pairs $\ldots$

- Let

$$\hat{P}_t(s'|s,a) = \frac{n_{s,a,s'}(t)}{1 \vee n_{s,a}(t)}$$

under let

$$\| \hat{P}_t(\cdot|s,a) - P(\cdot|s,a) \|_1 = \sum_{s'} |\hat{P}(s'|s,a) - P(s'|s,a)|$$

(P: true P)

We want to find an f s.t. $P \in \mathcal{C}_{t,\delta}$

w.p. $\geq 1-\delta$ $\forall t \geq 1$

$$\mathcal{C}_{t,\delta} = \left\{ \tilde{P}(\cdot|s,a) : \| \tilde{P}(\cdot|s,a) - \hat{P}_t(\cdot|s,a) \|_1 \leq f(n_{s,a}(t)) \quad \forall s,a \right\}$$

Noting that $\hat{P}_t(\cdot|s,a)$ is an estimate obtained from $n_{s,a}(t)$ i.i.d. transitions from (s,a), let's denote $\hat{P}_t(\cdot|s,a)$ by $\overline{P}_{n_{s,a}(t)}(\cdot|s,a)$ for convenience by abusing notation.

Thus, $P \in C_{t,\delta} \quad \forall t \geq 1 \quad$ w.p. $\geq 1-\delta$

if $P(\cdot|s,a) \in$

$$\left\{ \tilde{P}(\cdot|s,a) : \| \tilde{P}(\cdot|s,a) - \overline{P}_n(s,a) \|_1 \leq f(n) \right\}$$

$\forall s,a,t \quad$ w.p. $\geq 1-\delta$

Now appealing to the discussion at the beginning of these notes, we choose

$$f(n) = \sqrt{\frac{2}{n}\left(S\log 2 + \log \frac{SA\,n(n+1)}{\delta}\right)}$$

since $\sum_{n\geq 1}\sum_{s,a} \frac{\delta}{n(n+1)SA} = \delta,$

where S is the cardinality of the states and A is the cardinality of actions